

Cyber tAPPs Week 4 (class 6 and 7)

Instructions: Work on problem 1 on your own to apply the concepts you learned in the pre-reading material to solve problems. Then discuss your code with your team members and present one solution as a team. Follow the same process for problems. Swap your team's solution with another team for feedback.

Orientation and navigation

Problem 1

A student runs:

```
ls /var/log > logs.txt >> backup.txt
```

The file logs.txt is empty, but backup.txt contains the log listing.

- Why did this happen?
- Rewrite the correct command so **both files** contain the log listing.

Problem 2 – Parsons Puzzle: File Navigation

Rearrange these commands so they correctly:

1. Move into reports/,
2. List all files,
3. Save the list into list.txt.

A. ls > list.txt

B. cd reports

C. pwd

Problem 3

A student types:

```
cat secure.txt
```

The shell replies:

```
cat: secure.txt: No such file or directory
```

Give two possible reasons for this error.

Suggest how the student can fix it.

Problem 4

A system admin wants to:

1. Create a directory called logs/.
2. Inside it, create a file errors.log.
3. Append the word ALERT into errors.log.

Write the sequence of commands (no more than 3 lines).

Problem 5

A student asks:

“If I am already in /, what happens when I run cd ..?”

Explain the behaviour clearly and why.

Input/Output and Shell Fundamentals

Problem 1

A student writes the following C code:

```
FILE *fp = fopen('data.txt', r);
```

It does not compile.

- 1) Identify **all errors** in this line.
- 2) Rewrite the correct code.

Problem 2

Rearrange the following lines to form a correct C program that:

1. Prompts the user for a filename,
2. Opens it,
3. Reads it line by line safely,
4. Prints the number of lines.

A. `fgets(buffer, sizeof(buffer), fp) != NULL`

B. `int count = 0;`

C. `printf("Enter filename: ");`
D. `FILE *fp = fopen(filename, "r");`
E. `while ( ... ) { count++; }`
F. `printf("Lines = %d\n", count);`

Problem 3

A student types:

```
echo $USER is logged in
```

The output is:

```
Student20 is logged in
```

But when they type:

```
echo "$USER is logged in"
```

It shows:

```
Student20 is logged in
```

Explain why the first command might break if the username contains spaces, and why quotes fix it.

Problem 4

You have a file `security.log`.

You need to:

1. Extract all lines containing "ALERT",
2. Count how many there are,
3. Save the number into a file called `alert_count.txt`.

Write the one-line command that does this.

Problem 5

A student tries to run:

```
ls && pwd || echo "done"
```

They are confused why "done" does not always appear.

- a) Explain **how && and || operators work** in this command.

- b) Rewrite a version where "done" **always** runs at the end.

C Programming Basics

Problem 1

```
char buf[8];  
strcpy(buf, "ProgrammingSD");
```

What bug occurs?

Rewrite using a safe alternative.

Problem 2

```
char arr[10];  
for (int i = 0; i <= 10; i++) {  
    arr[i] = 'A';  
}
```

- What happens?
- How can this be exploited?

Problem 3

Write a C program that:

1. Prompts for two strings.
2. If not equal, concatenates them.
3. Copies into a third string.
4. Prints all lengths.

Using Vim and the C Toolchain

Problem 1

Makefile:

run:

```
./program
```

run:

`./other`

Why does make run only execute `./other`? Rewrite the correct version so **both run**.

Problem 2

Student runs:

```
gcc hello.c
```

```
./hello
```

The output is:

```
Error: ./hello: No such file or directory
```

- a) Why?
- b) What flag must they add to gcc?

Problem 3

Write a Makefile with:

- hello target compiling `hello.c`.
- run target executing `./hello`.
- clean target deleting the executable